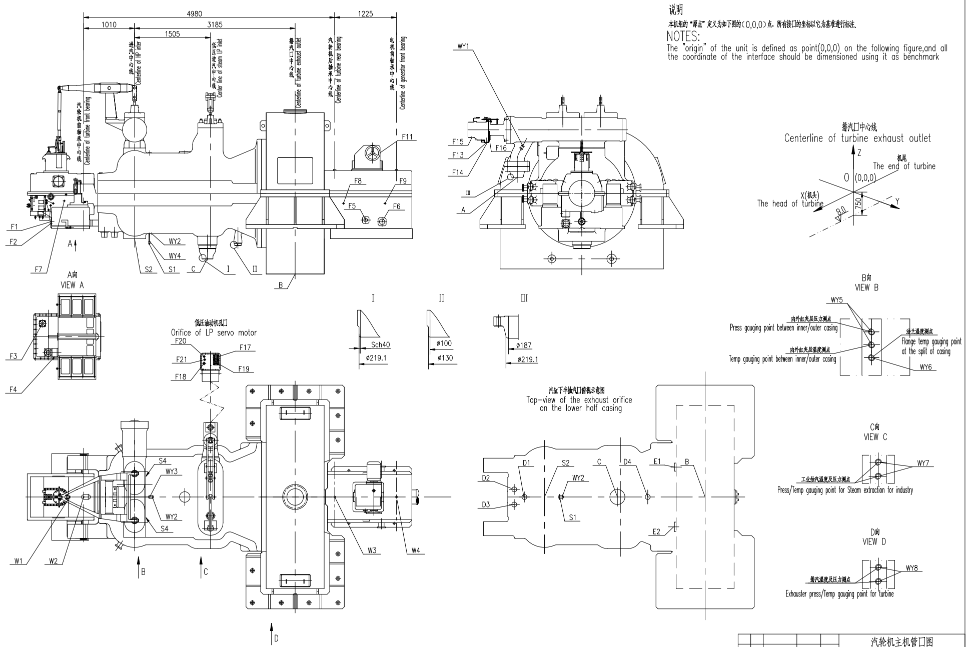
## 说明

本机组的“原点”定义为如下图的（0,0,0）点，所有接口的坐标以它为基准进行标注.

## NOTES:

The "origin" of the unit is defined as point(0,0,0) on the following figure,and all the coordinate of the interface should be dimensioned using it as benchmark


4980
1010
3185
1505
1225
汽机前轴承中心线 Centerline of turbine front bearing
高压进汽中心线 Centerline of turbine HP inlet
排汽口中心线 Centerline of turbine exhaust outlet
后轴承中心线 Centerline of turbine rear bearing
发电机前轴承中心线 Centerline of generator front bearing
排汽口中心线
Centerline of turbine exhaust outlet
X(机头) The head of turbine
Z
机尾 The end of turbine
O (0,0,0)
Y
250
80
F1
F2
F3
F4
F5
F6
F7
F8
F9
F11
F13
F14
F15
F16
F17
F18
F19
F20
F21
WY1
WY2
WY3
WY4
WY5
WY6
WY7
WY8
W1
W2
W3
W4
S1
S2
S4
A
B
C
D
D1
D2
D3
D4
E1
E2
I
II
III
A向 VIEW A
B向 VIEW B
C向 VIEW C
D向 VIEW D
低压油动机接口
Orifice of LP servo motor
Sch40
ø219.1
ø100
ø130
ø187
ø219.1
汽缸下半排汽口俯视示意图
Top-view of the exhaust orifice on the lower half casing
内外缸夹层压力测点
Press gauging point between inner/outer casing
内外缸夹层温度测点
Temp gauging point between inner/outer casing
法兰温度测点
Flange temp gauging point at the split of casing
工业抽汽温度及压力测点
Press/Temp gauging point for Steam extraction for industry
排汽温度及压力测点
Exhauster press/Temp gauging point for turbine

| 设 计 | |
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|---|---|---|---|---|---|
| 标记 | 处数 | 更改文件号 | 签 字 | 日期 | 汽轮机主机管口图<br>Orifice drawing of turbine proper |
| 设 计 | | 审 定 | | | 0-0000-586C-0000-00 重 量(Kg) |
| 校 对 | | 日 期 | 20260115 | | 青岛捷能汽轮机集团股份有限公司<br>QINGDAO JIENENG STEAM TURBINE GROUP CO., LTD. |
| 审 核 | | 共2页 | 第1页 | | |

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### 主要管口表
### Orifice list for main pipe I/O

| 代 号<br>Code | 名 称<br>Name | 形式/规格<br>Model/Size | 数量<br>Quantity | X | Y | Z | 连接尺寸标准<br>Joint size standards | 备 注<br>Remark |
|---|---|---|---|---|---|---|---|---|
| 油 路 参考 外部油管路、 前轴承座、后轴承座、盘车装置<br>Oil I/O Reference drawing: External oil line,Front bearing housing,Rear bearing housing,Turning gear device | | | | | | | | |
| F1 | 润滑油进口 lubrication oil inlet | 法兰PN0.6,DN50 flange | 1 | 4840 | 0 | -560 | | |
| F2 | 回油口 return oil | 法兰PN0.6DN100 flange | 1 | 4850 | -315 | -600 | | |
| F3 | 主油泵进口 main bump inlet | 法兰PN0.6DN65 flange | 1 | 5140 | 260 | -380 | | |
| F4 | 主油泵出口 main bump outlet | 法兰PN4.0,DN65 flange | 1 | 4910 | -410 | -400 | | |
| F5 | 润滑油进口 lubrication oil inlet | 法兰PN0.6,DN65 flange | 2 | -1425 | ±460 | -530 | | 左右对称symmetry |
| F6 | 回油口 return oil | 法兰PN0.6,DN125 flange | 2 | -1750 | ±605 | -565 | | 左右对称symmetry |
| F7 | 1#瓦顶轴进口 jacking oil inlet | 管接头G6/M10×1 pipe joint | 1 | 4580 | 410 | -150 | | |
| F8 | 2#瓦顶轴进口 jacking oil inlet | 管接头G6/M10×1 pipe joint | 1 | -985 | 460 | -200 | | |
| F9 | 3#瓦顶轴进口 jacking oil inlet | 管接头G6/M10×1 pipe joint | 1 | -1810 | 460 | -200 | | |
| | | | | | | | | |
| F11 | 盘车润滑油 lubri oil to turning gear | 管接头G8/M12×1.5 pipe joint | 1 | -1721 | 0 | 802 | | |
| | | | | | | | | |
| F13 | 去主汽门保安油 safy oil to MSV | 管接头 G14/M18×1.5 pipe joint | 1 | 3185 | -1892.5 | 1005 | | |
| F14 | 去主汽门高压油 HP oil to MSV | 管接头 G10/M14×1.5 pipe joint | 1 | 3185 | -1852.5 | 1045 | | |
| F15 | 回油口 return oil | 管接头 G18/M22×1.5 pipe joint | 1 | 3185 | -2183 | 1103 | | |
| F16 | 回油口 return oil | 管接头 G8/M12×1.5 pipe joint | 1 | 3185 | -1830 | 1065 | | |
| F17 | 接主油泵入口管路 to main bump inlet | 法兰 Flangle | 1 | 1570 | -1661 | 460 | | |
| F18 | 接泄油管路 To oil tank | 管接头 G28/M33x2 pipe joint | 1 | 1808 | -1541 | 470 | | |
| F19 | 接高压油管路 High press oil | 法兰 Flangle | 1 | 1570 | -1561 | 460 | | |
| F20 | 接脉冲油管路 Pulse oil | 管接头G18/M22x1.5 pipe joint | 1 | 1805 | -1676 | 470 | | |
| F21 | 接保安油 safy oil | 管接头G18/M22×1.5 pipe joint | 1 | 1795 | -1616 | 470 | | |

| 代 号<br>Code | 名 称<br>Name | 形式/规格<br>Model/Size | 数量<br>Quantity | X<br>ΔX | Y<br>ΔY | Z<br>ΔZ | 连接尺寸/标准<br>Joint size/standards | 备 注<br>Remark |
|---|---|---|---|---|---|---|---|---|
| 汽 路 参考 热力系统图、汽缸电站图<br>Steam I/O Reference drawing:Thermidynamic system drawing,Casing | | | | | | | | |
| A | 进汽口<br>steam inlets | 法兰DN200-16<br>flange | 1 | 3185<br>11.65 | -1280<br>-7.12 | 315<br>1.75 | | |
| | | | | | | | | |
| D1 | 接D4<br>Connect with D4 | DN100-1.6 | 1 | 3585<br>13.26 | 0<br>0 | -870<br>-3.90 | ø108x4.5 | |
| D2 | 接均压箱<br>Connected with Surger tank | DN100-1.6 | 1 | 3785<br>14.15 | -150<br>-0.67 | -870<br>-3.90 | ø108x4.5 | |
| D3 | 接轴封冷却器<br>Connected with the gland cooler | DN100-1.6 | 1 | 3785<br>14.15 | +150<br>0.67 | -870<br>-3.90 | ø108x4.5 | |
| D4 | 接D1<br>Connect with D1 | DN100-1.6 | 1 | 1135<br>3.17 | 0<br>0 | -1040<br>-4.69 | | |
| C | 工业抽汽<br>Adjustable extraction | DN200-1.6 | 1 | 1740<br>5.26 | 0<br>0 | -1303<br>-3.46 | | |
| S1 | 调节级后疏水<br>Drain before governing stage | 等径三通T(S) 20-Ⅱ-Sch80 | 1 | 2895 | 0 | -1000 | | |
| S2 | 汽缸疏水<br>Drain of casing | 管接头G18/M22x1.5 | 1 | 3185 | 0 | -900 | | |
| | | | | | | | | |
| S4 | 蒸汽室盖疏水<br>Drain of steam lid | 管接头G12/M20x1.5 | 2 | 2965 | ±375 | 1455 | | 左右对称symmetry |
| E1 | 接均压箱<br>Connected with Surger tank | 法兰DN65-0.6<br>flange | 1 | 608<br>1.07 | -593<br>-1.97 | -1300<br>-4.3 | ø108x4.5 | |
| E2 | 接轴封冷却器<br>Connected with the gland cooler | 法兰DN80-0.6<br>flange | 1 | 608<br>1.07 | +593<br>1.97 | -1300<br>-4.3 | ø108x4.5 | |
| B | 排汽口<br>Exhaust outlet | 1150x3640 | 1 | 0<br>0 | 0<br>0 | -1600<br>-0.31 | | 左右对称symmetry |

**注：ΔX、ΔY、ΔZ为汽缸各管口在X、Y、Z方向上的热膨胀值，法兰联接的X、Y、Z值为法兰结合面尺寸.**

**Note: ΔX、ΔY、ΔZ are thermal expansion for orifices of casing in X、Y、Z direction,**
**the X、Y、Z values of flage connecting are dimensions of flange adjacent plan.**

**未备注压力单位均为MPa（No Remarks pressure unit herein is MPa.）**

### 汽轮机主要温度及压力测点表
### Turbine Main temp/press gauging point table

| 序号 | 用 途 | Usage | 管 口 Orifice |
|---|---|---|---|
| W1 | 汽机推力轴承回油温度测点 | Temp gauging point forreturn oil of turbine thrust bearing | M27x2 |
| W2 | 汽机前轴承回油温度测点 | Temp gauging point for return oil of turbine front bearing | M27x2 |
| W3 | 汽机后轴承回油温度测点 | Temp gauging point for return oil of turbine rear bearing | M27x2 |
| W4 | 发电机前轴承回油温度测点 | Temp gauging point for return oil of generator front bearing | M27x2 |
| WY1 | 汽机高压进汽压力/温度测点 | HP inlet press/Temp gauging point for turbine | 接管/M33x2 Joint pipe |
| WY2 | 外缸壁温度测点 | Temp gauging point for casing wall | M20x1.5 |
| WY3 | 调节级后蒸汽温度测点 | Temp gauging point behind control stage | M20x1.5 |
| WY4 | 调节级后蒸汽压力测点 | Press gauging point behind control stage | 接管 Joint pipe |
| WY5 | 内外缸夹层蒸汽压力/温度测点 | Press/Temp gauging point between inner/outer casing | 接管/M20x1.5 Joint pipe |
| WY6 | 汽缸中分面法兰温度测点 | Flange temp gauging point at the split of casing | M20x1.5 |
| WY7 | 汽机排汽压力/温度测点 | Exhauster press/Temp gauging point for turbine | 接管/M20x1.5 Joint pipe |
| WY8 | 工业抽汽压力/温度测点 | Press/Temp gauging point for Steam extraction for industry | 接管/M20x1.5 Joint pipe |

### 汽缸各管口力与力矩限定值
### Limited force and moment for orifices of casing

| 管口公称直径DN | 允许的力（N） | | | | 允许的力矩（N.m） | | | |
|---|---|---|---|---|---|---|---|---|
| | X向 | Y向 | Z向 | 合力 | X向 | Y向 | Z向 | 合力矩 |
| DN50 | 500 | 1000 | 1250 | 1677 | 750 | 375 | 375 | 919 |
| DN65 | 650 | 1300 | 1625 | 2180 | 975 | 487 | 487 | 1194 |
| DN80 | 800 | 1600 | 2000 | 2683 | 1200 | 600 | 600 | 1470 |
| DN100 | 1000 | 2000 | 2500 | 3354 | 1500 | 750 | 750 | 1837 |
| DN125 | 1250 | 2500 | 3125 | 4193 | 1875 | 940 | 940 | 2298 |
| DN150 | 1500 | 3000 | 3750 | 5031 | 2250 | 1125 | 1125 | 2756 |
| DN200 | 2000 | 4000 | 5000 | 6708 | 3000 | 1500 | 1500 | 3674 |
| DN250 | 2167 | 4334 | 5417 | 7268 | 3250 | 1625 | 1625 | 3980 |
| DN300 | 2333 | 4667 | 5833 | 7826 | 3500 | 1750 | 1750 | 4287 |
| DN350 | 2500 | 5000 | 6250 | 8385 | 3750 | 1875 | 1875 | 4593 |
| DN400 | 2667 | 5334 | 6667 | 8945 | 4000 | 2000 | 2000 | 4899 |
| DN450 | 2833 | 5667 | 7083 | 9503 | 4250 | 2125 | 2125 | 5205 |
| DN500 | 3000 | 6000 | 7500 | 10062 | 4500 | 2250 | 2250 | 5511 |
| DN700 | 3667 | 7333 | 9167 | 12298 | 5500 | 2750 | 2750 | 6736 |
| DN800 | 4000 | 8000 | 10000 | 13416 | 6000 | 3000 | 3000 | 7348 |

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| 标记 | 处数 | 更改文件号 | 签 字 | 日期 | 汽轮机主机管口图<br>Orifice drawing of turbine proper |
| 设 计 | | 审 定 | | | 0-0000-586C-0000-00 重 量(Kg) |
| 校 对 | | 日 期 | 20260115 | | 青岛捷能汽轮机集团股份有限公司<br>QINGDAO JIENENG STEAM TURBINE GROUP CO., LTD. |
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